

11

TACTICAL CREATIVITY AND DECISION MAKING IN SPORT

Daniel Memmert and André Roca

Introduction

A crucial attribute of high-performance athletes is the ability to be surprising in their decision-making processes, thereby allowing them to deal more effectively with unique performance situations while making it hard for opponents to predict what they will do next. A current example of this is one of the best football players in the world, Lionel Messi, who is able to make decisions that are unexpected and more likely to set up his teammates' shots on goal. The importance of tactical creativity in sports seems to be increasing due to coaches' ability to collect more and more information about their opponents by means of game observation analysis (e.g. studying the individual and group tactical behaviour patterns of a team). Thus far, few researchers in the sport sciences have considered the importance of tactical creativity in sport. In contrast, the analysis of general and domain-unspecific creativity is a popular topic among psychologists (Milgram, 1990; Runco, 2007; Sternberg, 1999; Sternberg & Lubart, 1999; for a recent overview, see DeDreu, Nijstad, Baas, Wolsink, & Roskes, 2012). Over the last decade, creativity research has regained popularity, mostly due to some extraordinary findings in the field of modern functional neuroanatomy (Cabeza & Nyberg, 2000; Dietrich, 2004; Fink, Graif, & Neubauer, 2009).

In a general scientific context, Sternberg and Lubart (1999, p. 3) define creativity as "the ability to produce work that is both novel (i.e. original, unexpected) and appropriate (i.e. useful)". In sport science, motor and tactical creativity are discussed within the emergence of adaptive motor and cognitive solution processes occurring in complex environments (Moraru, Memmert, & van der Kamp, 2016). Motor creativity or creative motor actions refer to solutions that are (statistically) rare and thus original (Memmert & Perl, 2009; Simonton, 2003; for a discussion, see Orth, van der Kamp, Memmert, & Savelsbergh, 2017). In team and racket

sports, the importance of tactical creativity has been highlighted (Memmert, 2013, 2015a, 2015b). The process of being creative is distinct from the so-called best solution approaches (e.g. tactical intelligence, game playing ability, game sense; see Grehaigine, Wallian, & Godbout, 2005; Light, 2004; Mitchell, Oslin, & Griffin, 2006; Roca, Ford, & Williams, 2013; Rossi, Fry, McNeill, & Tan, 2007). Tactical creativity is understood to be the adequate, useful, varying, rare, and flexible production of tactical response patterns (Memmert & Roth, 2007). This difference between expert decision making and creativity may be based on the theoretical distinction between ‘divergent thinking’ and ‘convergent thinking’ (Guilford, 1967). Sternberg and Lubart (1999, p. 3) define divergent thinking as the unusualness, innovativeness, statistical rareness, or uniqueness of solutions in a given task on a behavioural level. In team and racket sports, Memmert (2015a) defines tactical creativity as a divergent tactical thinking process to allow the generation of a variety of solutions in specific individual, group, and team situations that are useful, surprising, seldom, and therefore original.

The distinct meaning of tactical creativity is elucidated when you consider divergent thinking ability in the entire process of human decision making (see Sternberg & Lubart, 1999). First, a situation is anticipated and perceived based on past experiences (memory); then attention is paid to specific targets of the situation; and finally a range of ideas is collected (creativity), with one of them being selected (intelligence) as the solution to the game situation presented. Tactical creativity is a relatively complex process where a number of other factors may be positively or negatively associated with it, such as motivation, training, and expertise. For example, most original solutions are highly context specific and embodied in emotions, and have a motivational background (Memmert, Hüttermann, & Orliczek, 2013).

In this chapter, we present an overview of literature on tactical creativity and decision making in team sports. We start by contextualising the value of ‘tactical creativity’ and how this term is operationally defined and theoretically considered. We discuss the different cognitive processes (e.g. anticipation, perception, attention, game intelligence, and working memory) underpinning tactical creativity in sports. Additional factors that appear linked with tactical creativity are then discussed, including age, expertise, level of training, and motivation. Finally, we close the chapter by considering potential directions for future research in the field of creativity in sport.

Cognitions and divergent tactical thinking in team sports

In the field of cognitive psychology, recent developments suggest the existence of different subprocesses during decision making (e.g. Memmert, 2015b; see Figure 11.1). These include anticipation, perception, attention, game intelligence, tactical creativity, and working memory. In this section, we use these to describe a classification framework for categorising research studies according to their focus on tactical creativity in sports.

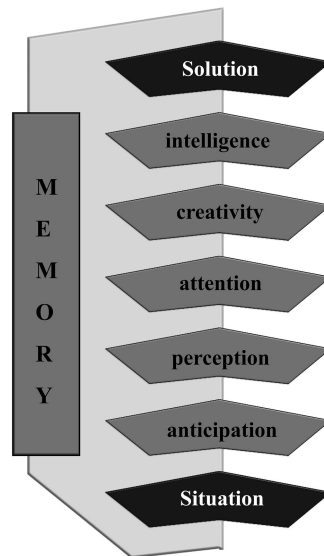


FIGURE 11.1 An overview of the central cognitive performance factors that underlie all actions in team and racket sports (adopted from Memmert, 2015b).

Anticipation and tactical creativity

An extensive body of published research has been gathered over the last few decades to identify the key factors underpinning anticipation in sport. The importance of several perceptual-cognitive skills is highlighted such as postural cue usage, pattern recognition, and the use of situational probabilities. Although both anticipation and tactical creativity are considered crucial in many different dynamic, time-constrained tasks across a range of sports (see Abernethy, Farrow, Gorman, & Mann, 2012; Memmert, 2015a), there have been few, if any, well-controlled efforts to examine how anticipation informs, interacts with, and facilitates creative performance solutions. It is predicted that the ability of performers to recognise advance cues in the environment, which enables them to anticipate what will happen next, will likely place them in a better position to make decisions that are unexpected and less easily predicted by their opponent(s), thereby facilitating tactical creativity.

Perception and tactical creativity

Roca, Ford, and Memmert (2018) attempted to examine creativity in the decision-making and visual search behaviours of skilled soccer players during simulated 11-a-side attacking match play situations. Video clips were occluded at a key moment and participants were required to play a ball in response to each situation presented. The creativity of their solutions on the task was measured using

three observation criteria, originality, flexibility, and fluency of decisions, which were subsequently used to categorise players into most-creative and least-creative groups. Differences between these groups in decision making were underpinned by differences in visual search strategy, recorded during the task using a portable eye-movement registration system. The most-creative players employed a broader attentional focus, characterised by more fixations of shorter duration in a different sequential order and towards more informative locations of the display compared with less creative players. Additionally, following an exploratory analysis of the visual search data, it was identified that most-creative players spent more time fixating on attackers in threatening positions (i.e. a teammate in a dangerous position that could lead to a goal scoring opportunity if they received a pass) and detected these players earlier on in the attacking play.

In a follow-up study, using the same task paradigm, Roca, Memmert, and Ford (2017) attempted to determine how performers translate the information scanned and perceived from the visual display into appropriate creative solutions. Retrospective verbal reports of thinking were collected alongside visual search data across a large sample of skilled soccer players. The most-creative group of players generated a greater number of verbal report statements related to assessment of the current situation and planning of future actions when compared with the least-creative players. These findings highlight that creative performance is underpinned by different underlying perceptual-cognitive processes, which appear to be crucial in facilitating more creative solutions. Findings support previous research that examined the perceptual processes underpinning anticipation and decision-making expertise during similar dynamic, open-play game situations (e.g. Roca, Ford, McRobert, & Williams, 2011; Vaeyens, Lenoir, Williams, Mazyn, & Philippaerts, 2007). This finding highlights the association between domain-specific expertise and creativity. The ability of creative performers to use more effective visual search strategies to identify key visual information in the environment and predict likely outcomes in a given situation will likely place them in a better position to plan and select more original and flexible decisions. In addition to advance visual cues, pattern recognition, or knowledge of situational probabilities, creative thinking is seen as a further cognitive skill of expert performance (Memmert, 2011).

Attention and tactical creativity

Recently, researchers have supported the view that fewer instructions by coaches during game play lead to a wider breadth of attention and facilitate greater improvements in tactical creativity (Memmert, 2015a). In a six-month longitudinal study Memmert (2007) compared an attention-broadening training programme (less instructions by the coaches) with a second treatment group that focussed on the training of a narrow breadth of attention (many instructions by the coaches). Creative performance was measured by a real-world sport-specific creativity task (Memmert, 2010) and only improved

following attention-broadening training. These findings support previous research on attentional narrowing (Dewing & Battye, 1971; Friedman, Fishbach, Förster, & Werth, 2003; Martindale, 1999; for a review, see Kasof, 1997). A narrow breadth of attention limits the amount of stimuli and information that can be perceived and reduces the potential of discovering unique and original solutions (Memmert & Furley, 2007; Roca, Memmert, & Ford, 2015; Roca, et al., 2017, 2018). A wide breadth of attention makes it possible to associate different stimuli that may initially appear to be irrelevant (Furley, Memmert, & Heller, 2010). The findings by Memmert (2007) highlight the fact that using an attention-broadening programme can promote the development of creativity in children. More recently, researchers have scrutinised the relationship between participants' control of visual attention and their creativity. Results revealed an influence of motivational states and negative expectancies as well as an impact on participants' ability to control their visual attention (for an overview, Hüttermann, Nerb, & Memmert, 2018). For example, different motivational conditions or regulatory motivational foci influence the tactical creativity of soccer players, which we discuss in the next section.

Tactical intelligence and tactical creativity

In a field-based study by Memmert and Roth (2007), the tactical creativity and tactical intelligence of children with an average age of seven years was determined at three measurement points (0 months (baseline), then at six and nine months, respectively). At consecutive measurement points, correlation coefficients of 0.36, 0.55, and 0.71 were obtained between divergent and convergent performance. These are comparable to those reported earlier in psychological studies (cf. Runco & Albert, 1986; Yamamoto, 1965). One of the assumptions of the common theory (discussed by Cropley, 1995) is confirmed; high intelligence is correlated with low to moderate levels of creativity, whilst higher levels of game intelligence are accompanied by higher tactical creativity.

Working memory and tactical creativity

The capacity and general function of working memory seem to be especially important for tactical creativity (for a review, see Furley & Memmert, 2010). Although an athlete's working memory capacity does not seem to influence divergent performance in team and racket sports (Furley & Memmert, 2015), it is nevertheless advisable that a trainer does not deliver too much information when delivering tactical instructions (Memmert, 2015a). Furley and Memmert (2012) demonstrated that the active content of working memory guides an athlete's focus of attention by biasing attention towards related objects in the visual field. In addition, working memory is active in controlling attention in a goal-directed manner to avoid distraction and interference amongst athletes (Furley & Memmert, 2012).

Other factors that appear linked with tactical creativity

To date, researchers have mostly analysed in relative isolation the importance of cognitive processes (e.g. anticipation, perception, attention, game intelligence, and working memory) responsible for producing creative solutions, especially in team and racket sports. Therefore, there is a lack of research examining mediating or moderating variables. To overcome this limitation, researchers should emphasise the importance of searching for complex correlation patterns between all these resources. Alongside research that illuminates the link between motivation, expertise, perception, and creativity, significant insight is expected from experimental examination of attention theories. Unconscious processes serve as an early selection mechanism, which favours useful or emotionally interesting information for further processing (Memmert, 2017b). In addition, we encourage research using the dynamical systems and ecological approaches, which offer a framework to examine how different constraints influence the emergence of motor and tactical creativity (Orth et al., 2017). By such approaches, it will be possible to examine multifaceted tasks in a complex environment in both a theoretical and practical way in order to come up with new ideas for practitioners (Dos Santos, Memmert, Sampaio, & Leite, 2016). To this end, we now consider research highlighting several variables that are important in the development of creativity (Memmert, 2017a).

Age

Many researchers have shown that creativity must be learned and developed early in life (for a review, see Milgram, 1990). Research from neuroscience suggests that stagnation in creativity development may occur after the age of eight years (Huttenlocher, 1990). Primary results from a cross-sectional design by Memmert (2011) supported the evidence from neuroscience and indicated that creativity should be cultivated early in a child's development. Comparisons between age constellations show that creativity of performance in children and adolescents (peer groups: 7, 10, and 13 years) does not develop linearly. From 7 to 10 years of age, considerable increases in tactical creativity are evident. This can be connected to the fact that the absolute number of synapses and the synapse density reaches its maximum in this age range (Casey, Tottenham, Liston, & Durston, 2005). By implication, training sessions with children in team and racket sports should focus on developing tactical creativity early in the training process.

Expertise

Skilled players with high scores on a sport-specific inattentional blindness task perform better than skilled players with low scores in regard to specific creative thinking abilities. In contrast, and in accordance with general creative thinking abilities, non-skilled players with lower scores on a general inattentional blindness task outperformed non-skilled players with higher scores (Memmert, 2010). In a longitudinal study of 195 talented football players in Germany aged

12 and 13 years, Memmert (2011) found that their performance on a test of general divergent thinking and a football-specific tactical creativity task did not change over a six-month period. Nevertheless, there was a slight tendency in younger athletes to improve their domain-specific tactical creativity. This finding raises the question of whether development of tactical creativity in skilled players stagnates at a certain age, which might warrant further investigation. However, there were large inter-individual differences in the level of improvement shown, which suggests they are not tightly constrained by age.

That expertise influences tactical creativity was revealed in a movement biography study by Memmert, Baker, and Bertsch (2010), in which 12 coaches carefully selected the most and least-creative players from their teams. For top team players in the highest national league or even in the national team (soccer, basketball, field hockey, and team handball), the number of hours of training activities makes the difference between more creative and less creative team sport players.

Training

The tactical creativity approach (TCA) by Memmert (2015a; see Figure 11.2) was proposed as a model for training and developing tactical creativity. The TCA focusses on seven methodological principles that foster tactical creativity in

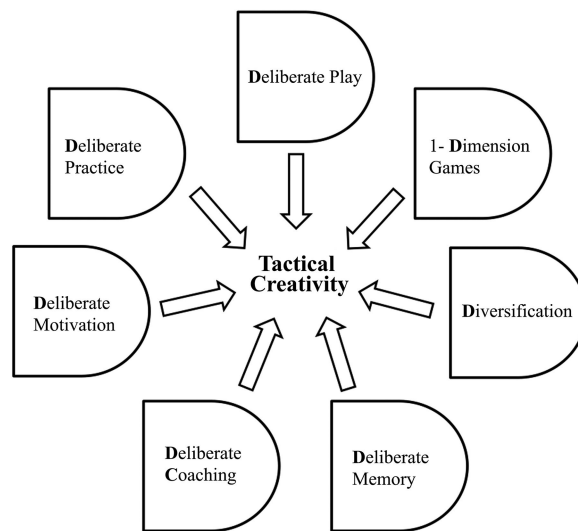


FIGURE 11.2 A theoretical framework – the Tactical Creativity Approach (TCA): the 7 D's fostering tactical creativity in team and racket sports. The order of the seven principles (starting with deliberate play going clockwise) indicates a chronological order from children and youth training to adolescence and adult training. While the first five principles are more suitable for younger age groups, all principles are useful for groups at an older age and should be integrated in training units (adopted from Memmert, 2017a).

team sports. All these principles (one-dimension games, diversification, deliberate practice, deliberate play, deliberate coaching, deliberate memory, deliberate motivation) are discussed on the basis of an empirical background.

Motivation

The nature of the instruction provided can have a direct influence on creative performance, which is demonstrated by current theoretical models and empirical results from social psychology (Hirt, Levine, McDonald, Melton, & Martin, 1997; Isen, 2000; Isen, Daubman, & Nowicki, 1987). Higgins (1997) recommended two modes of self-regulation in order to regulate pleasure and suffering and direct behaviour towards promotion or prevention targets. The first mode, characterised by a focus on accomplishments and aspirations, was labelled a promotion focus, while the second mode, characterised by a focus on safety and responsibilities, was called a prevention focus. A promotion focus is found when achievement of positive outcomes is perceived as pleasure and absence of those as suffering. In contrast, it is called a prevention focus if pleasure is identified as absence or successful avoidance of negative outcomes. Memmert et al. (2013) revealed that tactical performance also benefits from a promotion focus in sports. The authors demonstrated that soccer players were able to develop more original and flexible solutions in the promotion condition (e.g. instruction in a maze task: “Show the mouse the way to the cheese!”) than in the prevention condition (e.g. instruction in a maze task: “How can the mouse escape from the owl?”). The findings highlight the potential for coaches to use promotion focus instructions to enhance the divergent thinking of their athletes.

Conclusions and future directions

Creativity is a divergent thinking ability to produce adequate, original, and flexible solutions across different situational contexts (Sternberg & Lubart, 1999). It forms a key part of expert performance in sport (e.g., Kempe & Memmert, 2018) and other domains (e.g., Runco, 2007). The ability of high-performance athletes to be creative in their decision-making processes allows them to deal more effectively with unique performance situations while making it hard for opponents to predict what they do next. In this chapter, our goal was to provide a review of recent research on creativity as it relates to the development of skilled decision making. We provided an overview of the central cognitive processes responsible for the accomplishment of creative solutions, especially in team and racket sports, and examined other factors linked with tactical creativity (e.g. age and expertise levels, training, or motivation).

In future, researchers should focus on furthering our understanding of the dynamics and complex perceptual-cognitive processes underpinning creative decision making in sport (Memmert, 2015a, 2015b; Roca & Williams, 2016) as these appear to be crucial in facilitating the generation of more useful and surprising solutions. Further exploration and integration of multidimensional analysis, such

as eye-movement data; verbal report protocols; and behavioural analysis under diverse contextual, environmental, and emotional constraints, are clearly warranted (e.g. see Roca, Ford, McRobert, & Williams, 2013; Runswick, Roca, Williams, Bezodis, & North, 2017; Smith et al., 2016).

A further promising framework for explaining tactical creativity in a range of different sports is the concept of spontaneous self-organisation under constraints (Hristovski, Davids, & Araújo, 2006; Passos, Araújo, Davids, Gouveia, & Serpa, 2006). One aspect of the Nonlinear Dynamical Systems Theory link to tactical creativity is multistability (Hristovski, Davids, Araújo, & Passos, 2011), meaning that for a same set of constraints and for the same athlete's (personal) constraints, there may be more than one stable solution to the goal. Conversely, for an identically structured environment the type and the number of solutions may vary for different athletes. Moreover, a minuscule change in task constraints may bring about emergence or annihilation of different types and number of task solutions. As shown in Hristovski et al. (2006), this sensitivity is due to the fact that possibilities of action are scaled with respect to the abilities of the athlete. Hence, the position of the athlete along Guilford's creativity dimensions of fluency, flexibility, and originality will very much depend on these scaled athlete-environment relations.

Finally, we would like to encourage more people to attempt to capture and explore the multifaceted and complex nature of tactical creativity in sport, thereby helping advance our understanding of the specific processes underpinning sport-related creativity and how this quality is developed.

References

- Abernethy, B., Farrow, D., Gorman, A. D., & Mann, D. L. (2012). Anticipatory behavior and expert performance. In N. J. Hodges & A. M. Williams (Eds.), *Skill acquisition in sport: Research, theory and practice* (2nd ed., pp. 287–305). Abingdon, UK: Routledge.
- Cabeza, R., & Nyberg, L. (2000). Imaging cognition II: An empirical review of 275 PET and fMRI studies. *Journal of Cognitive Neuroscience*, 12, 1–47.
- Casey, B. J., Tottenham, N., Liston, C., & Durston, S. (2005). Imaging the developing brain: What have we learned about cognitive development? *Trends in Cognitive Sciences*, 9, 104–110.
- Cropley, A. (1995). Kreativität. In M. Amelang (Ed.), *Verhaltens- und Leistungsunterschiede. Themenbereich C. Serie VIII, Bd. 2* (pp. 329–373). Göttingen, Germany: Hogrefe.
- DeDreu, C. K., Nijstad, B. A., Baas, M., Wolsink, I., & Roskes, M. (2012). Working memory benefits creative insight, musical improvisation, and original ideation through maintained task-focused attention. *Personality and Social Psychology Bulletin*, 38, 656–669.
- Dewing, K., & Battye, G. (1971). Attention deployment and non-verbal fluency. *Journal of Personality and Social Psychology*, 17, 214–218.
- Dietrich, A. (2004). Neurocognitive mechanisms underlying the experience of flow. *Consciousness and Cognition*, 13, 746–761.
- Dos Santos, S. D. L., Memmert, D., Sampaio, J., & Leite, J. (2016). The spawns of creative behavior in team sports: A creativity developmental framework. *Frontiers in Psychology*, 7, 1282.

- Fink, A., Graif, B., & Neubauer, A. C. (2009). Brain correlates underlying creative thinking: EEG alpha activity in professional vs. novice dancers. *Neuroimage*, 46, 854–862.
- Friedman, R. S., Fishbach, A., Förster, J., & Werth, L. (2003). Attentional priming effects on creativity. *Creativity Research Journal*, 15, 277–286.
- Furley, P., & Memmert, D. (2010). The role of working memory in sports. *International Review of Sport and Exercise Psychology*, 3, 171–194.
- Furley, P., & Memmert, D. (2012). Working memory capacity as controlled attention in tactical decision making. *Journal of Sport and Exercise Psychology*, 34, 322–344.
- Furley, P., & Memmert, D. (2015). Creativity and working memory capacity in sports: Working memory capacity is not a limiting factor in creative decision making amongst skilled performers. *Frontiers in Psychology*, 6, 115.
- Furley, P., Memmert, D., & Heller, C. (2010). The dark side of visual awareness in sport: Inattention blindness in a real-world basketball task. *Attention, Perception, & Psychophysics*, 72, 1327–1337.
- Grehaigne, J. F., Wallian, N., & Godbout, P. (2005). Tactical-decision learning model and students' practices. *Physical Education & Sport Pedagogy*, 10, 255–269.
- Guilford, J. P. (1967). *The nature of human intelligence*. New York, NY: McGraw Hill.
- Higgins, E. T. (1997). Beyond pleasure and pain. *American Psychologist*, 52, 1280–1013.
- Hirt, E. R., Levine, G. M., McDonald, H. E., Melton, R. J., & Martin, L. L. (1997). The role of mood in quantitative and qualitative aspects of performance: Single or multiple mechanisms? *Journal of Experimental Social Psychology*, 33, 602–629.
- Hristovski, R., Davids, K., & Araújo, D. (2006). Affordance-controlled bifurcations of action patterns in martial arts. *Nonlinear Dynamics, Psychology, and Life Sciences*, 10, 409–444.
- Hristovski, R., Davids, K., Araujo, D., & Passos, P. (2011). Constraints-induced emergence of functional novelty in complex neurobiological systems: A basis for creativity in sport. *Nonlinear Dynamics, Psychology and Life Sciences*, 15, 175–206.
- Huttenlocher, P. R. (1990). Morphometric study of human cerebral cortex development. *Neuropsychologia*, 28, 517–527.
- Hüttermann, S., Nerb, J., & Memmert, D. (2018, in press). The role of regulatory focus and expectation on creative decision making. *Human Movement Science*.
- Isen, A. M. (2000). Positive affect and decision making. In M. Lewis & J. Haviland-Jones (Eds.), *Handbook of emotions* (2nd ed., pp. 417–435). New York, NY: Guilford.
- Isen, A. M., Daubman, K. A., & Nowicki, G. P. (1987). Positive affect facilitates creative problem solving. *Journal of Personality and Social Psychology*, 52, 1122–1131.
- Kasof, J. (1997). Creativity and breadth of attention. *Creativity Research Journal*, 10, 303–315.
- Kempe, M., & Memmert, D. (2018). “Good, better, creative”: The influence of creativity on goal scoring in elite soccer. *Journal of Sports Sciences*, 36, 2419–2423.
- Light, R. (2004). Coaches' experiences of games sense: Opportunities and challenges. *Physical Education & Sport Pedagogy*, 9, 115–131.
- Martindale, C. (1999). The biological basis of creativity. In R. J. Sternberg (Ed.), *Handbook of creativity* (pp. 137–152). Cambridge, UK: Cambridge University Press.
- Memmert, D. (2007). Can creativity be improved by an attention-broadening training program? An exploratory study focusing on team sports. *Creativity Research Journal*, 19, 1–12.
- Memmert, D. (2010). Testing of tactical performance in youth elite soccer. *Journal of Sports Science and Medicine*, 9, 199–205.
- Memmert, D. (2011). Creativity, expertise, and attention: Exploring their development and their relationships. *Journal of Sports Sciences*, 29, 93–104.

- Memmert, D. (2013). Tactical Creativity. In T. McGarry, P. O'Donoghue, & J. Sampaio (Eds.), *Routledge handbook of sports performance analysis* (pp. 297–308). Abingdon, UK: Routledge.
- Memmert, D. (2015a). *Teaching tactical creativity in sport: Research and practice*. Abingdon, UK: Routledge.
- Memmert, D. (2015b). Development of tactical creativity in sports. In J. Baker & D. Farrow (Eds.), *Routledge handbook of sport expertise* (pp. 363–372). Abingdon, UK: Routledge.
- Memmert, D. (2017a). Sports and creativity. In M. A. Runco & S. R. Pritzker (Eds.), *Encyclopedia of creativity* (Vol. 2, 3rd ed., pp. 373–378). San Diego, CA: Academic Press.
- Memmert, D. (2017b). Sports. In J. C. Kaufman, V. P. Glaveanu, & J. Baer, (Eds.), *Creativity across different domains* (pp. 479–491). Cambridge, MA: MIT Press.
- Memmert, D., Baker, J., & Bertsch, C. (2010). Play and practice in the development of sport-specific creativity in team ball sports. *High Ability Studies*, 21, 3–18.
- Memmert, D., & Furley, P. (2007). “I spy with my little eye!”: Breadth of attention, inattention blindness, and tactical decision making in team sports. *Journal of Sport & Exercise Psychology*, 29, 365–347.
- Memmert, D., Hüttermann, S., & Orliczek, J. (2013). Decide like Lionel Messi! The impact of regulatory focus on divergent thinking in sports. *Journal of Applied Social Psychology*, 43, 2163–2167.
- Memmert, D., & Perl, J. (2009). Analysis and simulation of creativity learning by means of artificial neural networks. *Human Movement Science*, 28, 263–282.
- Memmert, D., & Roth, K. (2007). The effects of non-specific and specific concepts on tactical creativity in team ball sports. *Journal of Sports Sciences*, 25, 1423–1432.
- Milgram, R. M. (1990). Creativity: An idea whose time has come and gone. In M. A. Runco & R. S. Albert (Eds.), *Theory of creativity* (pp. 215–233). Newbury Park, CA: Sage.
- Mitchell, S. A., Oslin, J. L., & Griffin, L. L. (2006). *Teaching sport concepts and skills: A tactical games approach* (2nd ed.). Champaign, IL: Human Kinetics.
- Moraru, A., Memmert, D., & van der Kamp, J. (2016). Motor creativity: The roles of attention breadth and working memory in a divergent doing task. *Journal of Cognitive Psychology*, 28, 856–867.
- Orth, D., van der Kamp, J., Memmert, D., & Savelsbergh, G. (2017). Creative motor actions as emerging from movement variability. *Frontiers in Psychology*, 8, 1903.
- Passos, P., Araújo, D., Davids, K., Gouveia, L., & Serpa, S. (2006). Interpersonal dynamics in sport: The role of artificial neural networks and 3-D analysis. *Behavior Research Methods*, 38, 683–691.
- Roca, A., Ford, P. R., McRobert, A. P., & Williams, A. M. (2011). Identifying the processes underpinning anticipation and decision-making in a dynamic time-constrained task. *Cognitive Processing*, 12, 301–310.
- Roca, A., Ford, P. R., McRobert, A. P., & Williams, A. M. (2013). Perceptual-cognitive skills and their interaction as a function of task constraints in soccer. *Journal of Sport & Exercise Psychology*, 35, 144–155.
- Roca, A., Ford, P. R., & Memmert, D. (2018). Creative decision making and visual search behavior in skilled soccer players. *PLOS ONE*, 13, e0199381.
- Roca, A., Ford, P. R., & Williams, A. M. (2013). The processes underlying ‘game intelligence’ skills in soccer players. In H. Nunome, B. Drust, & B. Dawson (Eds.), *Science and football VII* (pp. 255–260). London, UK: Routledge.
- Roca, A., Memmert, D., & Ford, P. R. (2015). Creativity, decision making and visual search in skilled soccer players. In J. Bangsbo & P. Krstrup (Eds.), *Proceedings of the 8th world congress on science and football* (p. 161). Copenhagen, Denmark: SL Grafik.

- Roca, A., Memmert, D., & Ford, P. R. (2017). Perceptual-cognitive processes underlying creativity in skilled soccer players. In *Proceedings of the 5th World Conference on Science and Soccer* (p. 64). Rennes, France: University Rennes 2.
- Roca, A., & Williams, A. M. (2016). Expertise and the interaction between different perceptual-cognitive skills: Implications for testing and training. *Frontiers in Psychology*, 7, 792.
- Rossi, T., Fry, J. M., McNeill, M., & Tan, C. W. K. (2007). The Games Concept Approach (GCA) as a mandated practice: Views of Singaporean teachers. *Sport, Education and Society*, 12, 93–111.
- Runco, M. A. (2007). *Creativity – theories and themes: Research, development, and practice*. Burlington, MA: Elsevier Academic Press.
- Runco, M. A., & Albert, R. S. (1986). The threshold theory regarding creativity and intelligence: An empirical test with gifted and nongifted children. *Creative Child & Adult Quarterly*, 11, 212–218.
- Runswick, O. R., Roca, A., Williams, A. M., Bezodis, N. E., & North, J. S. (2017). The effects of anxiety and situation-specific context on perceptual-motor skill: A multi-level investigation. *Psychological Research*, 82, 708–719.
- Simonton, D. K. (2003). Scientific creativity as constrained stochastic behavior: The integration of product, person, and process perspectives. *Psychological Bulletin*, 129, 475–494.
- Smith, M. R., Zeuwts, L., Lenoir, M., Hens, N., De Jong, L. M., & Coutts, A. J. (2016). Mental fatigue impairs soccer-specific decision-making skill. *Journal of Sports Sciences*, 34, 1297–1304.
- Sternberg, R. J. (Ed.) (1999). *Handbook of creativity*. Cambridge, UK: Cambridge University Press.
- Sternberg, R. J., & Lubart, T. I. (1999). The concept of creativity: Prospects and paradigms. In R. J. Sternberg (Ed.), *Handbook of creativity* (pp. 3–15). Cambridge, UK: Cambridge University Press.
- Vaeyens, R., Lenoir, M., Williams, A. M., Mazyn, L., & Philippaerts, R. M. (2007). Mechanisms underpinning successful decision making in skilled youth soccer players: An analysis of visual search behaviors. *Journal of Motor Behavior*, 39, 395–408.
- Yamamoto, K. (1965). Effects of restriction of range and test unreliability on correlation between measures of intelligence and creative thinking. *British Journal of Educational Psychology*, 35, 300–305.